CSE 225

Mid Exam

ECE, NSU

Duration: Exam Time (50 minutes) + Submission Preparation Time (15 minutes)

Total Marks: 50 (Answer all the Questions)

Guidelines

- All hand written contents should be submitted as a pdf file (in Google Classroom) and all the Answers must be submitted at once.
- The front page must include your name, student id, course id and section number and the remaining page(s) must include your student id and the page number.
- You must solve the exam problems independently and you will be fully responsible to maintain the *confidentiality* of your answers to avoid plagiarism.
- The submission preparation time (at the end of exam) is to make *pdf* of your hand written answers. Marks will be deducted proportionately for any delay in submission.

Problem 1. [10 Marks]
Fill in the array in *Figure 1* with your cell phone number. Re-draw the array to show each changes in the array [no code required] for sorting the values (in **descending** order) using ***Insertion*** sort :

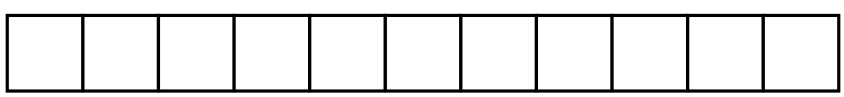

Figure 1

Problem 2. [15 Marks]
Write a new method for the **Unsorted List** (*Linked-List* based) class implementation to move the last node of the list to the beginning of the same list.

Problem 3. [10 Marks]

```
struct NodeType
{
ItemType info ;
NodeType* next ;
} ;
```

Consider the following **Unsorted List** (*Linked-List* based) created with the above node declaration. Fill in the info part of the nodes with the digits of your cell phone number (sequentially).

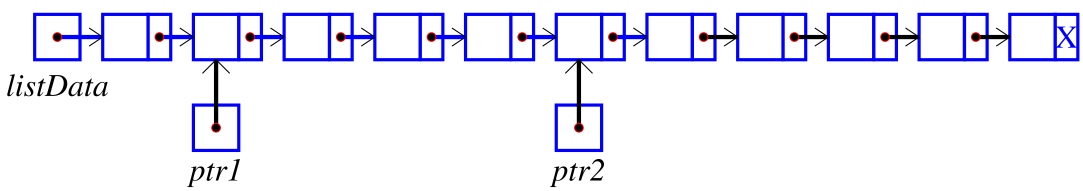

Figure 2

Draw *Figure 2* with cell phone digits & write down the output of the following expressions:

a) ptr2 → next → info

b) listData → next → next → info

c) listData → next == ptr1

d) ptr2 → next == NULL

e) listData → info == 0

Problem 4. [5 Marks]
Consider that, in your main method, you are using a **Unsorted List** (*Array* based) containing the values ranging from 1 to N (but in unsorted manner):

```
int N = list.GetLength(); //N contains the size of the UnsortedType List
for(int i = 0; i < N;i++)
    list.getItem(i);
```

If the above *for* loop takes 1 second to complete all the operations for a list of size 1000, how long would it take (approximately) to run on a list of size 10,000?

Problem 5. [10 Marks]
Consider a new method - **rearrange( )** for the **Unsorted List** (*Linked-List* based) class implementation. The rearrange( ) method re-organizes the nodes of the list. Consider the list in *Figure 2* where *listData* points to the first node of the list and each node contains a digit of your cell phone number. Draw the sequence of nodes after all the executions of **rearrange( )** method.

```
struct NodeType
{
    ItemType info;
    NodeType *next;
};
void rearrange()   //listData is the pointer to the first node
{   NodeType *p, *q;
    ItemType temp;
    if((listData == NULL) || (listData->next == NULL))
    {
        return;
    }
    p = listData;
    q = listData->next;
    while(q != NULL)
    {
         temp = p->info;
         p->info = q->info;
         q->info = temp;
         p = q->next;

         if(p != NULL)
         {
           p = p -> next;

           if(p != NULL)
            {
                q = p->next;
            }
            else
            {
                q = NULL;
            }
         }
         else
         {
            q = NULL;
         }
    }
}
```